BUSINESS ANALYTICS / MANAGEMENT REPORT

# Retail Sales Intelligence Platform

Global Superstore | 2011–2014

A data-led view of sales, profitability, customers, products and regional performance.



ANALYSIS BUILT FROM THE SUPPLIED CSV + PYTHON + SQL + POWER BI EVIDENCE

# Growth is strong; margin discipline is the next management question

Verified baseline recalculated from the supplied cleaned CSV.

| Total sales | Total profit | Profit margin | Orders | Customers |
|---|---|---|---|---|
| $12.64M | $1.47M | 11.61% | 25,035 | 1,590 |

- Sales grew from $2.26M in 2011 to $4.30M in 2014; profit rose each year.
- Technology leads category sales and profit; Central leads regional sales and profit; Consumer is the largest segment.
- Profitability varies sharply by sub-category, product and discount level; investigate weak-margin combinations before changing policy.
- 1,580 of 1,590 customers placed more than one distinct order in the full dataset (99.37% repeat-customer rate).

# The project tests how revenue converts into profit

Scope follows the supplied Business Problem Brief.

DECISION TO SUPPORT

## Where to focus

Product • customer • region • pricing

ANALYTICAL LENS

## Sales → profit

Volume • margin • mix • time

EVIDENCE STANDARD

## No causal leap

Patterns are not proven root causes

- Build a consistent baseline for sales, profit, distinct orders, customers and average order value.
- Compare performance across product/category, customer segment, geography and time.
- Identify loss-making products and examine discount and shipping-cost patterns.
- Translate validated findings into prioritized management actions and measurable monitoring.

# The CSV is the current calculation baseline; database reconciliation remains open

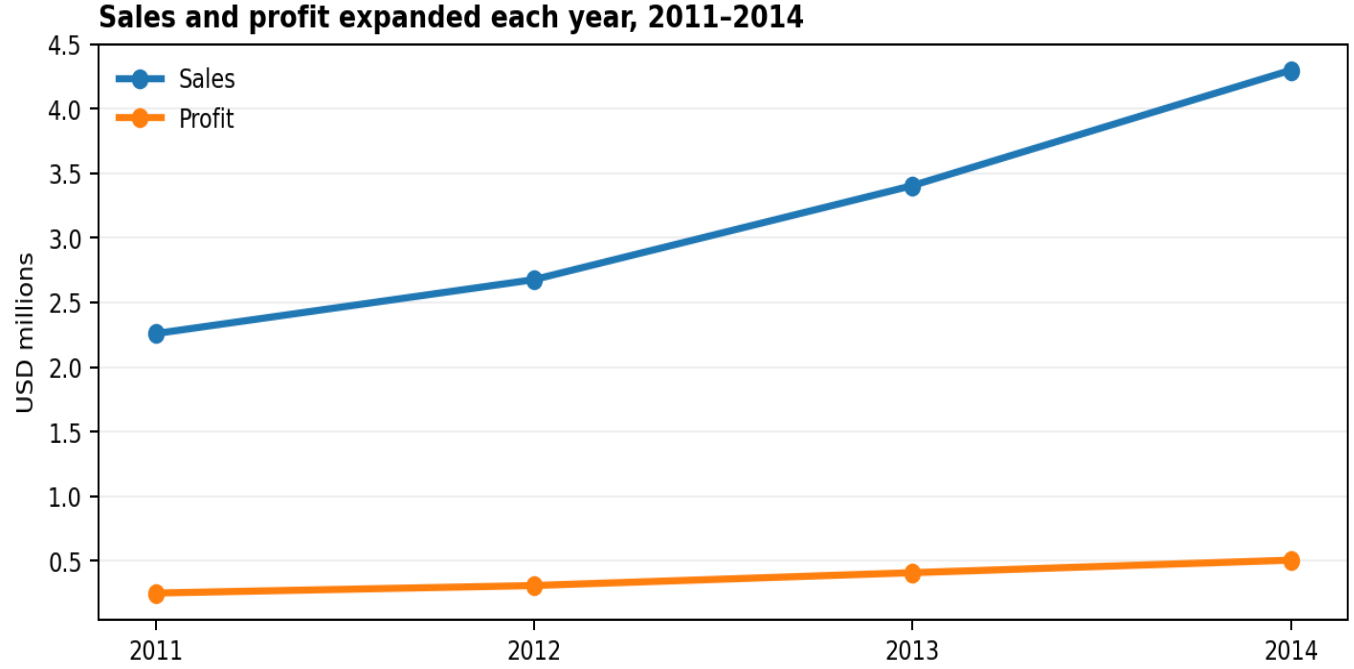
Source: supplied clean_global_superstore CSV and notebook.

| Rows / fields | Exact duplicate rows | Missing postal codes | Date coverage |
|---|---|---|---|
| 51,290 / 24 | 0 | 41,296<br>80.5% of records | 2011–2014 |

- No missing values were found in the other 23 columns of the supplied CSV.
- The notebook converts order/ship dates, checks duplicates, standardizes column names and exports a clean CSV.
- MySQL execution is not validated: the reported error 1146 says projectdata.global_superstore does not exist.
- The business brief flags differing totals between CSV and SQL screenshots; publish only one reconciled baseline.

# Sales and profit increased in every year of the observed period

Annual totals and year-over-year change calculated from the CSV.

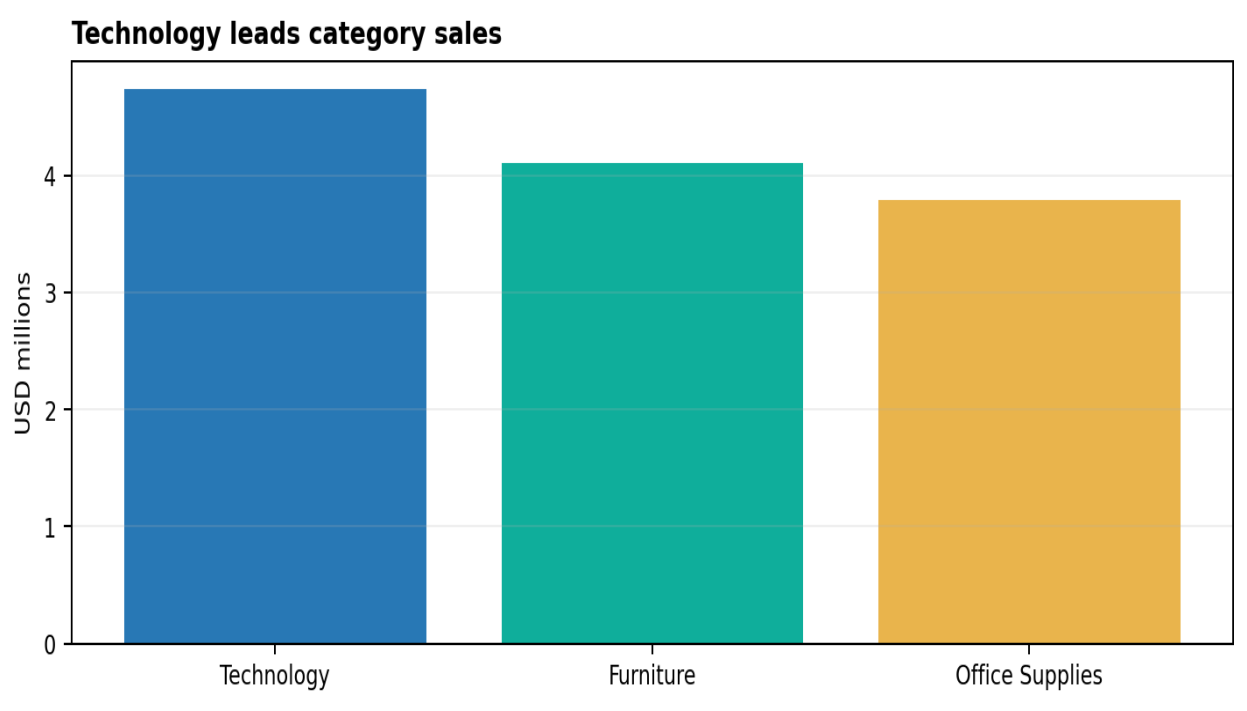



| Year | Sales | Profit | Sales YoY |
|---|---|---|---|
| 2011 | $2.26M | $249K | — |
| 2012 | $2.68M | $307K | 18.5% |
| 2013 | $3.41M | $407K | 27.2% |
| 2014 | $4.30M | $504K | 26.3% |

**2014 closed at $4.30M sales and $504.2K profit. Sales growth moderated slightly in 2014 versus 2013, while profit growth also eased.**

# Technology leads both revenue and profit; Furniture has the weakest category margin

Category totals aggregated from transaction lines.

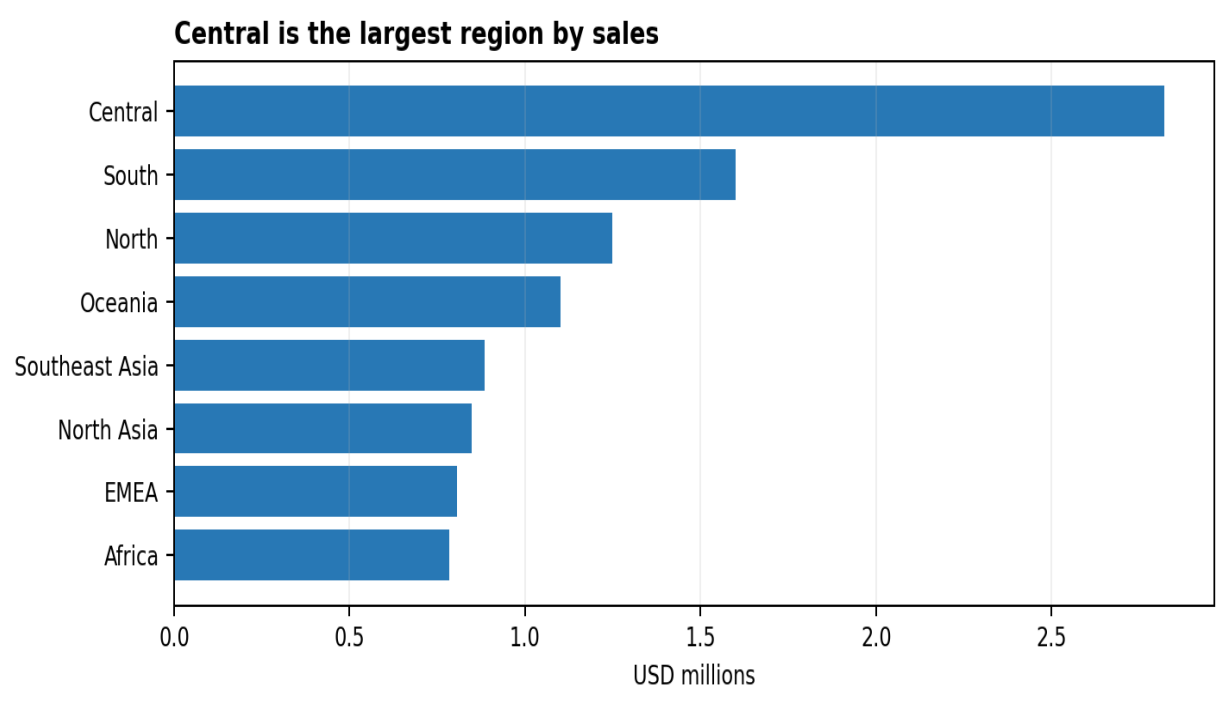



| Category | Sales | Profit | Margin |
|---|---|---|---|
| Technology | $4.74M | $664K | 13.99% |
| Furniture | $4.11M | $285K | 6.94% |
| Office Supplies | $3.79M | $518K | 13.69% |

**Technology margin: 13.99%. Furniture margin: 6.94%.**

# Central is the largest region by both sales and profit

Regional totals are descriptive and should be read alongside market and product mix.

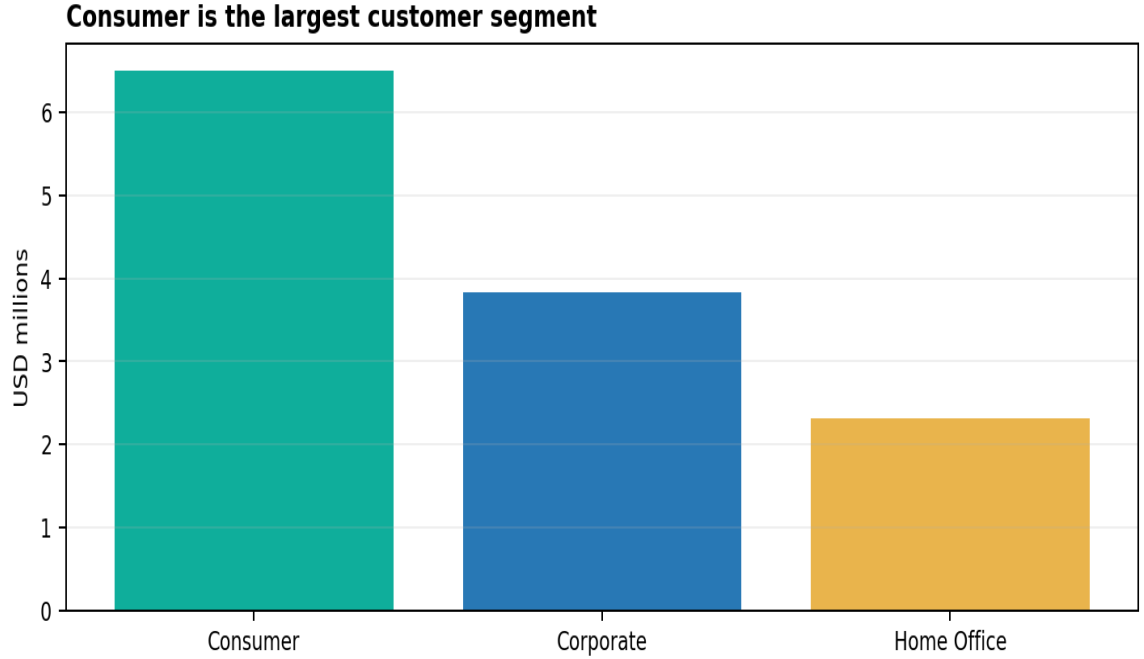



| Region | Sales | Profit | Margin |
|---|---|---|---|
| Central | $2.82M | $311K | 11.0% |
| South | $1.60M | $140K | 8.8% |
| North | $1.25M | $195K | 15.6% |
| Oceania | $1.10M | $120K | 10.9% |
| Southeast Asia | $0.88M | $18K | 2.0% |
| North Asia | $0.85M | $166K | 19.5% |
| EMEA | $0.81M | $44K | 5.4% |

**Southeast Asia generated $0.88M sales but only $17.9K profit (2.02% margin)—a clear investigation candidate.**

# Consumer contributes the largest segment; repeat rate is high under the stated definition

Repeat = more than one distinct Order_ID per Customer_ID across the full 2011–2014 file.




Repeat customers

**1,580**

Repeat rate

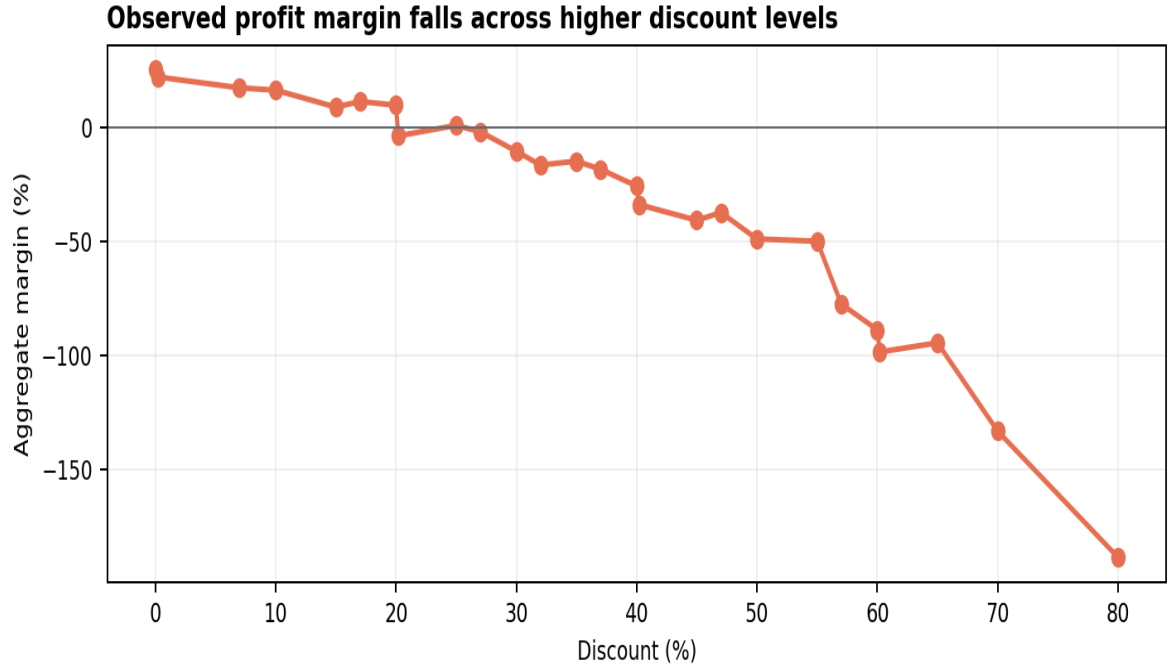
**99.37%**

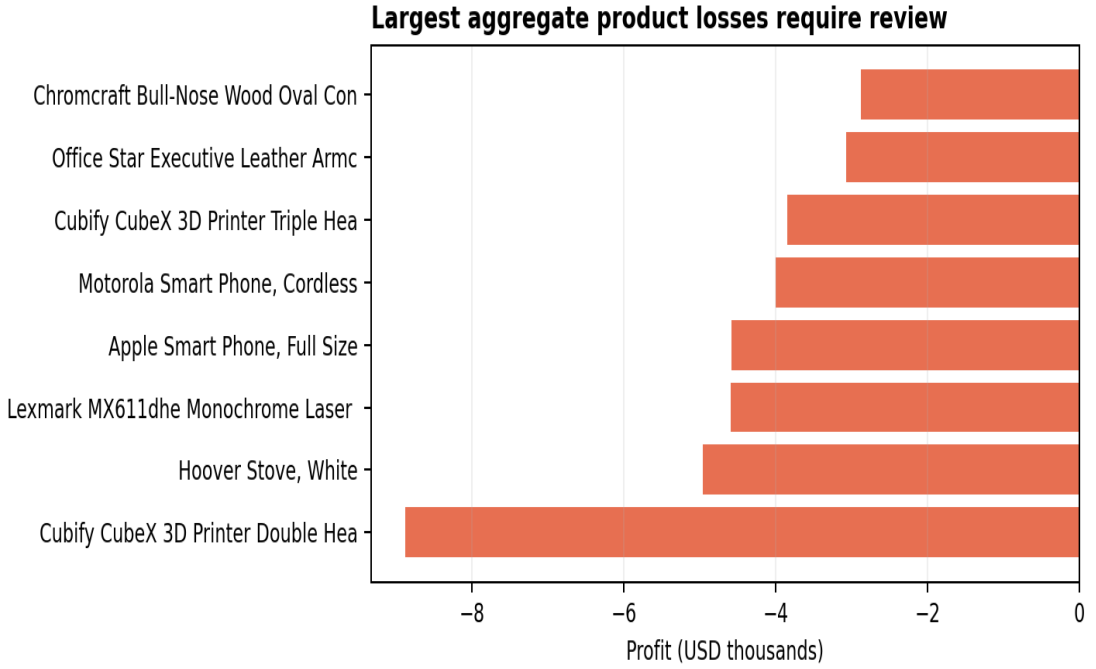
| Top sales customer | Sales | Orders |
|---|---|---|
| Tom Ashbrook | $35,668 | 25 |
| Greg Tran | $34,472 | 30 |
| Tamara Chand | $34,218 | 28 |
| Sean Miller | $31,125 | 21 |
| Bart Watters | $30,614 | 35 |

**Full-period repeat rate is not a cohort retention metric; calculate it by acquisition/first-order cohort before using it to evaluate retention programs.**

# Higher discount bands show weaker aggregate margins; several products are net loss-making

These are associations in grouped data, not proof of discount causation.




**Example: the 40% discount group recorded a -25.69% aggregate margin; the 70% group recorded -133.32%. Review sample sizes, product mix and transaction-level costs before setting discount rules.**

# Shipping modes differ in order volume, elapsed days and shipping cost

Average elapsed calendar days = Ship_Date minus Order_Date.

| Ship mode | Orders | Sales | Profit | Avg days | Shipping cost |
|---|---|---|---|---|---|
| First Class | 3,821 | $1.83M | $208K | 2.18 | $308K |
| Same Day | 1,347 | $0.67M | $76K | 0.04 | $116K |
| Second Class | 5,119 | $2.57M | $293K | 3.23 | $314K |
| Standard Class | 15,154 | $7.58M | $891K | 5.00 | $615K |

- Standard Class represents the largest order volume (15,154) and averages 5.00 elapsed days.
- Same Day averages 0.04 elapsed calendar days; date-only fields cannot measure intraday service time.
- Use mode-level comparisons for monitoring, not to infer service quality or cost causality.

# Prioritize margin visibility and data reconciliation before intervention

Actions are proposed; no financial benefit is claimed.

| Priority | Workstream | Monitor | Next step |
|---|---|---|---|
| 1 • High | Discount guardrails | Margin by discount band; loss lines | Test thresholds by product/category and validate mix. |
| 2 • High | Loss-product review | Product profit, margin, units | Review the 2,892 negative-profit products and top loss items. |
| 3 • High | Category/region drill-down | Category × region margin | Investigate Furniture and low-margin regional combinations. |
| 4 • High | Reconcile data model | Rows, sales, profit, orders, customers | Match Power BI and MySQL to the CSV baseline. |
| 5 • Medium | Customer cohort view | Repeat rate by cohort; profit/customer | Move beyond full-period repeat rate to cohort retention. |
| 6 • Medium | Shipping scorecard | Avg days; cost/order; profit | Monitor service and shipping economics by ship mode. |

# A trusted performance baseline is the first deliverable.

- CSV baseline has been recalculated and is internally consistent for the metrics reported.
- MySQL remains unexecuted because the referenced table is not found; import/rename and rerun the SQL checks.
- Power BI screenshots show rounded values and some measures that must be reconciled to the same CSV baseline.
- The dataset supports descriptive analysis, not causal proof of why discounts, shipping or product outcomes occur.

**RECOMMENDED NEXT STEP: reconcile the source table and Power BI model, then publish a single governed KPI baseline.**

# Four dashboard pages supplied for review

Screenshots are retained as supplied; visual values are not treated as independently validated unless reconciled to the CSV.

